HACKATHON ARCHITECTURAL BLUEPRINT

Self-Evolving AI Agent Networks

Autonomous Real-Time Global Supply Chain Operations & Predictive Market Sensing

Domain: Multi-Agent Systems (MAS)
Focus: Closed-Loop Orchestration
Format: System Whitepaper

> **Core Thesis:** Traditional Enterprise Resource Planning (ERP) operates as passive software requiring manual human coordination for every edge-case. This paradigm transforms global operations into an **autonomous multi-agent state machine** that anticipates macro shocks, executes bounded interventions, and continuously self-refines its decision heuristics.

01 CORE SYSTEM PILLARS

Self-Evolving Mechanics — Feedback Loops

Agents do not rely on static rules or frozen prompt weights. They capture structured trajectory logs, measure delta against actual delivery/cost metrics, and dynamically optimize prompt strategies and tool routings via meta-evaluators.

Decentralized Agent Network — MAS Topologies

Monolithic LLMs fail across expansive logistics. Specialized micro-agents—handling Procurement, Inventory, Routing, and Demand Sensing—interact over standardized inter-agent communication protocols to negotiate trade-offs.

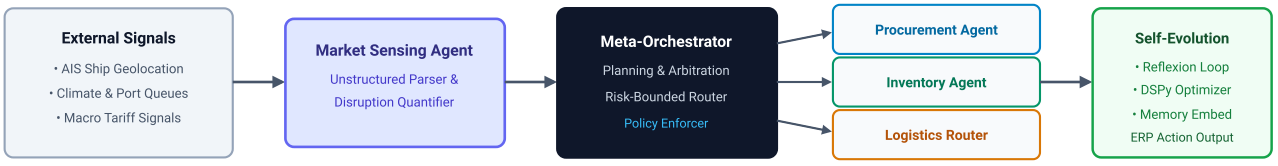
Autonomous Closed-Loop Execution — Deterministic Actions

Transits beyond passive alerts. When disruption probabilities exceed threshold boundaries, authorized sub-agents trigger API calls directly into operational ERPs (SAP/NetSuite) under bounded financial and operational guardrails.

Predictive Market Sensing — Multimodal Ingestion

Ingests real-time unstructured signals (AIS maritime telemetry, weather satellite radar, port labor dispute signals, fuel benchmarks) to forecast disruptions days before they manifest on internal balance sheets.

02 DISTRIBUTED AGENT TOPOLOGY & CONTROL FLOW



This decentralized architecture decouples detection from domain-specific resolution. The **Meta-Orchestrator** maintains invariant checks across the organization, preventing local optimization anomalies (e.g., procurement placing oversized bulk orders that violate working-capital limits).

03 SELF-EVOLUTION MECHANISMS & MEMORY FABRICS

MECHANISM	ALGORITHMIC IMPLEMENTATION	OPERATIONAL VALUE
Reflexion & Episodic Memory	Failure cases write traces to vector storage (pgvector). Agents retrieve retrospective "lessons learned" during planning stages.	Prevents recurring errors on unreliable shipping lanes or high-default vendors.
DSPy Teleprompter Optimization	Replaces manual prompt engineering with Bayesian or MIPRO optimizers targeting cost-deviation loss functions.	System self-adapts prompts dynamically as seasonal freight market volatilities shift.
Digital Twin Sandbox (Monte Carlo)	Agents simulate potential logistics routing policies across 10,000 synthetic shock permutations before live ERP execution.	Guarantees zero catastrophe-level decisions by stress-testing risk tails prior to release.
Tool Parameter Fine-Tuning	Direct Preference Optimization (DPO) on structured tool call schemas to refine precision of ERP write actions.	Eliminates schema errors and reduces token overhead by up to 45%.

04 HACKATHON IMPLEMENTATION STACK & GOVERNANCE

State Graph Engine — LangGraph / AutoGen

Utilize stateful, cyclic directed graphs. Enables multi-turn negotiation, back-propagation of execution errors, and deterministic state transitions.

Interoperability Layer — MCP / JSON-Schema

Implement Anthropic's Model Context Protocol (MCP) to provide clean interfaces for live vessel tracking, weather data, and mock SAP/NetSuite APIs.

Tiered Autonomy Rails — Safety Guardrail

Decisions under **$10,000** with risk factor < 0.15 execute autonomously. Deviations above thresholds escalate to Human-in-the-Loop approval cards.

Evaluation Metrics — Scoring Framework

Continuously score agent outputs against: Mean Absolute Scaled Error (MASE), OTIF (On-Time In-Full) delivery rates, and Carrying Cost Overhead.

05 LIVE INCIDENT LIFECYCLE: PANAMA CANAL TRANSIT RESTRICTION

Incident Scenario: Severe climate drought cuts Panama Canal daily transits by 40%, generating a 12-day bottleneck across the Central America maritime corridor.

PHASE	ACTIVE AGENT	AUTONOMOUS ACTION TAKEN
1. Detect	Market Sensing Agent	Ingests maritime notice; computes regional lead-time penalty (+8 days) on scheduled cargo.
2. Assess	Inventory Agent	Flags Tier-1 factory in Dallas: assembly lines will experience stockouts within 96 hours.
3. Arbitrate	Meta-Orchestrator	Calculates factory downtime risk ($85,000/day) vs. expedited logistics cost premiums.
4. Execute	Procurement & Logistics	Triggers emergency PO to certified Mexican supplier; secures border freight slot via API.
5. Evolve	Evolutionary Loop	Saves trajectory vector: updates regional risk multiplier for future Latin America transits.